What moves an open AI-edit detector on technical prose: controlled edits, one feature at a time

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Abstract

Reading a score table suggests what to change in a text to lower its AI-edit score. We tested five such suggestions as controlled edits and three of them did not hold. We scored 62 paragraphs from four of our robotics papers with Pangram's open EditLens model. Joining two sentences with a semicolon raised the score of 92 % of the paragraphs, by +0.047 on average. Appending a short verdict sentence lowered it by 0.015, although verdict words had looked like a cause of high scores. Shuffling the sentences, changing "we" to "I" and appending a sentence of the same paper did nothing. Rewriting paragraphs against the sentence-level scores lowered the four paper means from 0.211, 0.176, 0.202 and 0.233 to 0.153, 0.160, 0.189 and 0.207. Pre-2022 human papers score 0.053 to 0.095, so none of the four reached that range.

1 Introduction

Our four papers pass the slopscore and lmscan checks of the repository, and alphaXiv, which runs Pangram's detector, still flags them as AI-written. We wanted to know what in the text the detector reacts to, so that we could write the next paper without it.

We first read a table of 469 sentences from the four papers and compared the score contribution of each sentence with its features. Semicolons, contrast framing, advice, verdict words and colon set-ups went with higher scores, and "we", short sentences and citations went with lower ones. From this table we wrote rule 78 of the writing rules in road-to-robotics. A feature that goes with a high score can travel with something else, such as a long list of numbers. This paper changes one feature at a time and scores the result.

EditLens [5] takes a text and regresses how much an AI edited it, as a score between 0 and 1. Its authors trained it on pairs of a source text and an AI-edited version of it, from reviews, news, stories and web articles, edited with prompts such as "make it more descriptive" and "fix any mistakes". Our texts are experimental robotics papers, so we did not expect the same features to matter. The controlled edits below test that.

2 Setup

The scorer is `pangram/editlens_Llama-3.2-3B`, the open model, on a Jetson Orin NX at 25 W. The production detector that flagged the papers is a different and stronger model, and we did not run it. The model is licensed for non-commercial use and we used it on our own four papers only.

The texts are the rendered drafts of four papers: a register of re-tested robot-learning claims, a JEPA world model on a low-cost arm, reward models and a stopping rule, and chunking tricks for a VLA on an edge GPU. Citations are stripped and the model lower-cases its input. Paragraphs of at least 40 words are scored on their own, since shorter ones are noisy. There are 62 of them, in 469 sentences. As a human reference we scored three robot-learning papers from before November 2022. Their paper means are 0.053, 0.055 and 0.095.

An edit changes one thing in every paragraph that contains it, and we report the mean change in the paragraph score with a 95 % bootstrap interval over paragraphs. An edit that changes only case and spacing gives exactly 0.000, because the model normalizes both. We removed semicolons and colons from the texts before this test, so for these two we could only test adding them.

3 Results

3.1 Controlled edits

The semicolon and the two closers raise the score, and the verdict sentence lowers it. Advice, the sentence about the rig, a real sentence of the same paper and the pronoun change do nothing, so adding length or changing the pronoun does not move the score. Shuffling does nothing either. Dropping the parentheticals, splitting at "and" and "but", and masking numbers lower the score. Keeping half of a paragraph lowers it most.

*Code, result files and the checks that regenerate every number: `github.com/ravediamond/paper-editlens-sensitivity`. The texts under test belong to four other papers of the author; their rewrite history is in `repro/rewrite-loop/`.

edit	paragraphs changed	change in score (95 % CI)
join sentence pairs with a semicolon	62	+0.047 (+0.039 to +0.056)
append a generic closer, “The result holds on this rig.”	62	+0.017 (+0.012 to +0.022)
append a contrast closer, “a limit of the setting, not of the method”	62	+0.011 (+0.006 to +0.017)
append advice, “Check the simple baseline first.”	62	+0.001 (−0.004 to +0.005)
append a verdict, “The row fails.”	62	−0.015 (−0.020 to −0.011)
append a sentence about the rig, “We ran this on the Jetson Orin NX.”	62	−0.001 (−0.007 to +0.004)
append a real sentence from another paragraph of the same paper	62	+0.003 (−0.005 to +0.011)
change “we” and “our” to “I” and “my”	52	+0.001 (−0.004 to +0.006)
shuffle the sentences of a paragraph	62	+0.000 (−0.012 to +0.012)
drop the parentheticals	27	−0.018 (−0.033 to −0.005)
split at “, and” and “, but”	28	−0.014 (−0.020 to −0.007)
replace every number with “many”	55	−0.021 (−0.030 to −0.012)
keep only the first half of the sentences	62	−0.029 (−0.044 to −0.015)

Table 1: Controlled edits on the original paragraphs. A positive change means the edited paragraph scores higher, which is worse for us.

feature	correlation table	controlled edit
semicolons	+0.028	+0.047, holds
contrast framing	+0.029	+0.011, holds and is smaller
advice or imperatives	+0.021	+0.001, does not hold
verdict words	+0.020	−0.015, reversed
“we” or “our” doing something	−0.020	+0.001, does not hold

Table 2: The sentence-level correlations we first used as rules, against the controlled edit of the same feature.

3.2 What the earlier table got right

Two of five features held. The verdict result is the one we did not expect. In the papers a verdict word sits in sentences that also carry a number, a bar and a citation, which may be what raised the score in the table. We did not test that. A verdict sentence appended on its own does not raise it.

3.3 The rewrite loop

We rewrote paragraphs from the highest-scoring sentences down. Each round scored a few candidate rewrites with the placeholders of the paper filled in, and kept the lowest one that preserved every number and claim. In the register paper one sentence moved to another paragraph and one that repeated another was dropped. There were 90 candidates in 50 paragraph-rounds. 50 of the candidates scored lower than the text they replaced, and the best candidate of a round improved the paragraph in 64 % of the rounds, by 0.032 on average.

paper	paragraphs	mean before	mean after
register of re-tested claims	23	0.211	0.153
JEPA world model	22	0.176	0.160
reward models and a stopping rule	21	0.202	0.189
chunking tricks on an edge GPU	18	0.233	0.207

Table 3: Word-weighted mean paragraph score before and after the rewrite.

change	paragraph	before	after
name the works in place of “papers propose several uses”	JEPA introduction	0.315	0.164
split one list sentence into plain statements	register, cold reruns	0.253	0.139
open with the AUROC example and give the class letter after it	register, classes B and C	0.253	0.145

Table 4: The three largest drops in one paragraph, by the change that caused them.

The largest single drops came from three changes, listed in Table 4. Merging sentences into longer ones with “so” or “which means” raised a limitations paragraph from 0.275 to 0.365 in the first variant, and the other two variants did the same. Paragraphs made of numbers, bars and verdicts stayed above the human range whatever we did.

3.4 A lexicon of words

We also tried to turn the sentence scores into a list of words. A lexicon built on three papers predicts the sentence effects of the fourth with a correlation of 0.116, and the 95th percentile of shuffled labels is 0.096. Deleting single words in the 20 highest paragraphs (2,042 deletions) agrees in sign with the sentence source for 48 of 80 words. Only 19 words pass our strict test. Words that name a method for talking about itself (protocol, rule, committed, reports, CI) go with higher scores, and plain task words and units go with lower ones. The effect of one word is small, since the mean deletion changes a paragraph by +0.013 and the largest by 0.157. The list is a tie-breaker between synonyms.

4 Limitations

The model is the open 3B one, and the detector that flagged the papers is another. We did not run the production model, so none of these effects is shown to carry over to it. Running it on the four papers costs about 3.2 dollars for each full pass at the listed API rate, and we did not spend it.

We drafted the four papers with Claude Code, and Claude Code wrote the rewrites in this paper too. The

author has not yet read the rewrites paragraph by paragraph. The open EditLens model scores a paragraph low when it sees little AI editing in it. We do not read the score as who wrote the paragraph.

Table 1 changes one feature per edit. A real rewrite changes several at once. We have not run the test that applies the rewrite moves of Section 3.3 one at a time to the same paragraphs. That test would show whether their effects add up. All 62 paragraphs come from the four papers of one author, on one SO-101 arm and one Jetson Orin NX. The five appended sentences are our own examples, and other wording could behave differently.

5 Related work

EditLens [5] introduces the continuous score and reports binary and ternary detection results, and we use its open 3B model. Its authors state that the model has a non-zero error rate and can cause harm when used to accuse someone. The AI Polish study [4], which EditLens evaluates on, shows that binary detectors flag lightly polished text as AI. Work on paraphrasers and humanizers [2, 1, 3] studies how they degrade the detection of AI text. We read only the descriptions of these in the EditLens paper and not the works themselves. We start from human-directed drafts of technical prose and ask which edits move an open model's score, and we report the edits that did nothing.

6 Conclusion

Of the five features we had read off a correlation table, a semicolon and a contrast closer raised the score, and the other three did not, so test a suspected feature as an edit before it becomes a writing rule. In our rewrites, paragraphs with named things and plain "we ran X on Y" sentences scored lower than paragraphs of procedure words, and result paragraphs stayed above the human range.

References

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